A new polymath proposal: explaining identities for irreducible polynomials

Terence Tao · 28 December 2015

https://terrytao.wordpress.com/2015/12/28/a-new-polymath-proposal-explaining-identities-for-irreducible-polynomials/

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Over on the polymath blog, I've posted (on behalf of Dinesh Thakur) a new polymath proposal, which is to explain some numerically observed identities involving the irreducible polynomials P in the polynomial ring $\mathbf{F}_2[t]$ over the finite field of characteristic two, the simplest of which is

$$\sum_P \frac{1}{1+P} = 0$$

(expanded in terms of Taylor series in $u = 1/t$). Comments on the problem should be placed in the polymath blog post; if there is enough interest, we can start a formal polymath project on it.

Comments (7)

Gil Kalai · 28 December 2015

This looks like a very nice project! If it will be developed into a full project this will be (I think) the first time that we have more than one polymath project running in parallel and this is very welcomed! Right now polymath10 devoted to the Erdos-Rado sunflower conjecture is running in my blog; Here is the link to the first post. It is quite possible that yet another polymath project will take place over a third blog in the coming months.

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>
> **Gil Kalai** · 29 December 2015
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> One more general polymath news: Here is the link of a mathoverflow question asking for polymath proposals http://mathoverflow.net/questions/219638/proposals-for-polymath-projects

Anonymous · 29 December 2015

It seems to involve the Carlitz zeta function. Is there any introductory material on that topic?

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>
> **Anonymous** · 29 December 2015
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> Also, Prof. Tao, is it a good idea to make a "reference request" on mathoverflow about Carlitz zeta functions so that we could have more reference?
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> >
> > **David Speyer** · 30 December 2015
> >
> > Goss's book is a good quick reference.
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> > >
> > >
> > > **Anonymous** · 30 December 2015
> > >
> > > so it seems done, right?

John Mangual · 29 December 2015

At first I thought of Turán's Sieve, but then I learned the irreducible polynomials can be counted by inclusion-exclusion: http://math.stackexchange.com/questions/40811/number-of-monic-irreducible-polynomials-of-degree-p-over-finite-fields
What else...? Plug in $t = 0, 1, \frac{1\pm\sqrt{-3}}{2}$ etc or something very close.